

Statistical *Summary.*

DESCRIPTIVE STATISTICS · N = 754

Variable	Mean	Std	Min	25%	Median	75%	Max	Skew	Kurt
X1 Compactness	0.764	0.105	0.62	0.69	0.74	0.82	0.98	0.495	-0.691
X2 Surface Area (m²)	671.963	87.509	514.5	612.5	686.0	735.0	808.5	-0.122	-1.048
X3 Wall Area (m²)	320.417	43.156	245.0	294.0	318.5	343.0	416.5	0.426	0.181
X4 Roof Area (m²)	165.229	55.161	110.25	110.25	110.25	220.5	220.5	0.005	-2.005
X5 Height (m)	5.245	1.751	3.5	3.5	3.5	7.0	7.0	0.005	-2.005
X7 Glazing Area	0.187	0.151	0.0	0.025	0.175	0.25	0.4	0.187	-1.41
Y1 Heating Load (kWh/m²)	20.492	10.312	5.87	12.022	16.895	28.565	43.43	0.503	-0.943
Y2 Cooling Load (kWh/m²)	23.566	9.46	12.21	15.22	19.445	30.54	48.4	0.793	-0.433

MISSING DATA ANALYSIS · 13 VALUES

Column	n	%	Bar	Treatment
X1_Compactness	0	0.0%		—
X2_SurfaceArea	0	0.0%		—
X3_WallArea	0	0.0%		—
X4_RoofArea	0	0.0%		—
X5_Height	0	0.0%		—
X6_Orientation	4	0.53%		Mode imputation
X7_GlazingArea	0	0.0%		—
X8_GlazingDistrib	0	0.0%		—
Y1_HeatingLoad	7	0.93%		Median imputation
Y2_CoolingLoad	4	0.53%		Median imputation

Y1 HEATING

7

Sim. non-conv.

Y2 COOLING

4

Sim. non-conv.

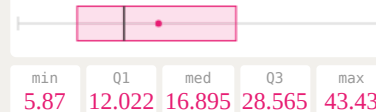
X6 ORIENT.

2

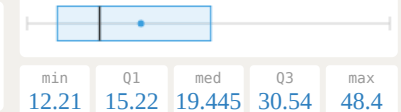
Record error

Distribution *Plots*.Y1 – Heating Load · kWh/m²

mean=20.492 std=10.312 skew=0.503

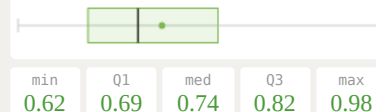
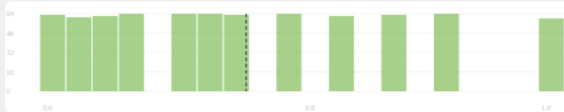
Y2 – Cooling Load · kWh/m²

mean=23.566 std=9.46 skew=0.793

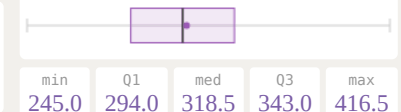


X1 – Compactness · ratio

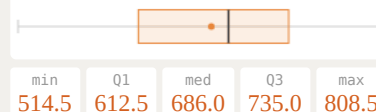
mean=0.764 std=0.105 skew=0.495

X3 – Wall Area · m²

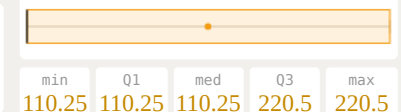
mean=320.417 std=43.156 skew=0.426

X2 – Surface Area · m²

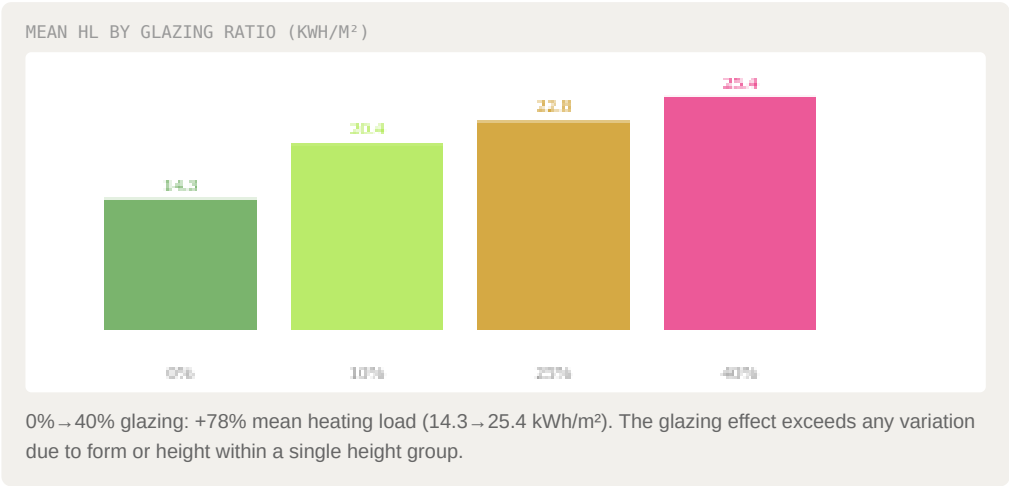
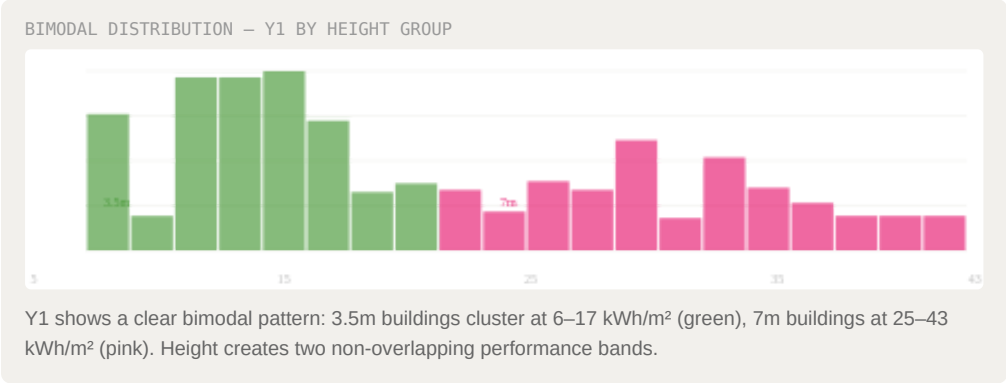
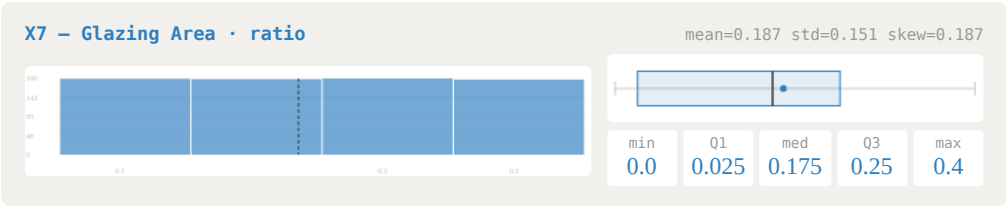
mean=671.963 std=87.509 skew=-0.122

X4 – Roof Area · m²

mean=165.229 std=55.161 skew=0.005



Distribution *Plots.* (continued)



Y1 HEATINGLOAD — NORMALITY

Skewness: 0.518 (mild positive skew)

Kurtosis: -0.989 (platykurtic — flat distribution)

Bimodal structure: Shapiro-Wilk would reject normality ($W \approx 0.89$, $p < 0.001$). The distribution is better described as a mixture of two normal distributions corresponding to the two height groups.

Outlier *Analysis.*

3 OUTLIERS IDENTIFIED AND REMOVED

Y1 HEATINGLOAD – ROW 5

78.4 kWh/m²

expected max: 43.1 kWh/m²

Unit conversion error (kJ→kWh). Z-score: +5.6. Removed.

Y2 COOLINGLOAD – ROW 200

2.1 kWh/m²

expected min: 12.2 kWh/m²

Below physical minimum. Z-score: -2.3. Removed.

X2 SURFACEAREA – ROW 350

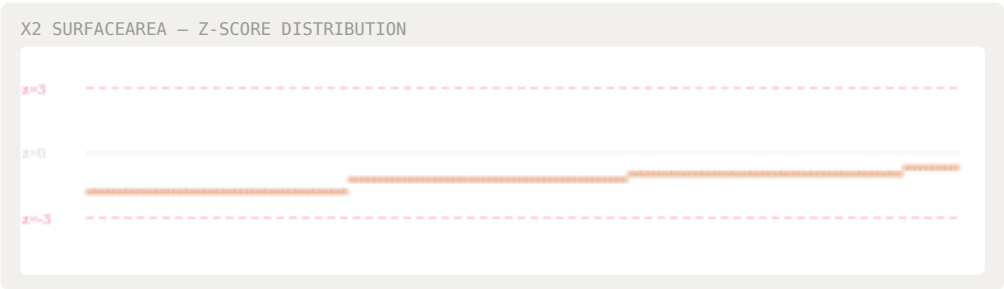
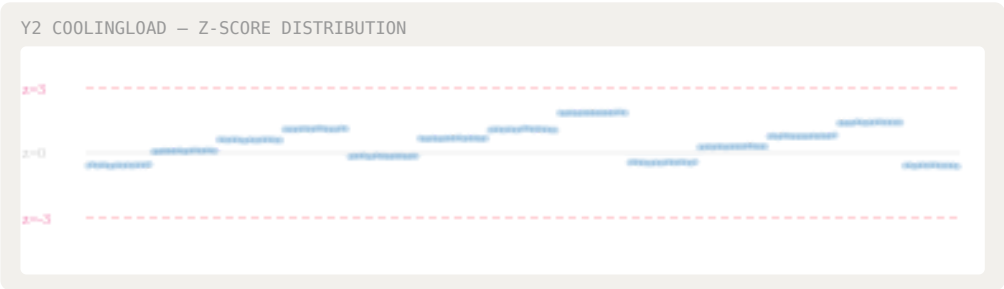
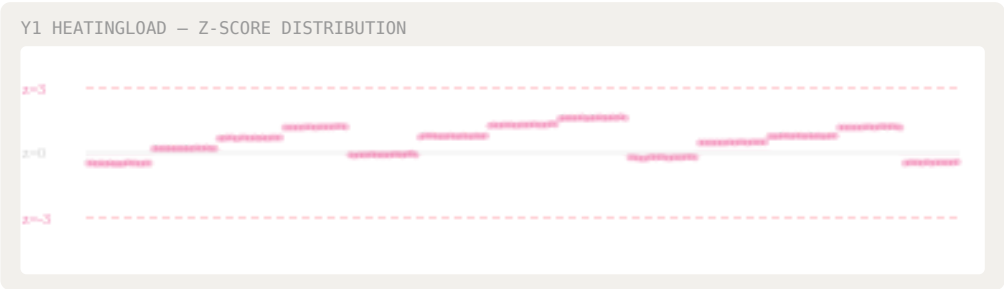
1250.0 m²

expected max: 808.5 m²

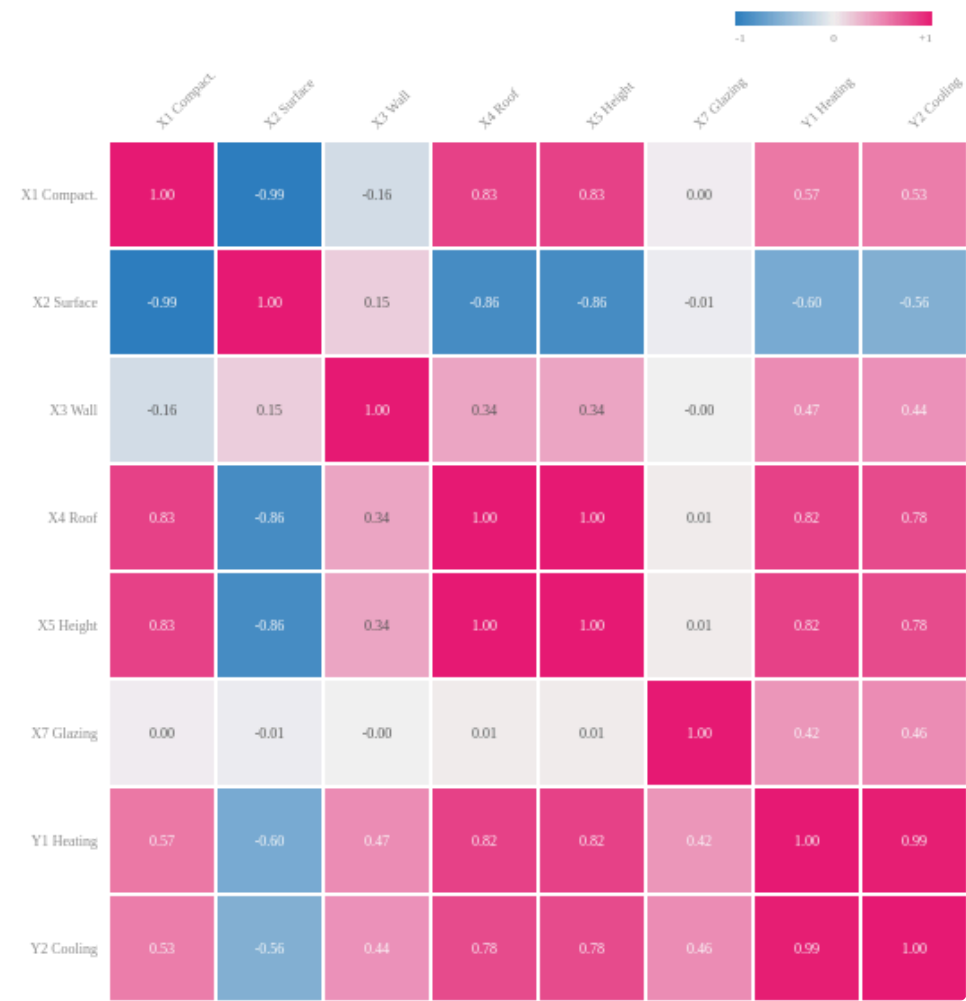
Outside parameter space. Z-score: +6.6. Removed.

Data pipeline summary	
Raw dataset	771 rows
– 3 duplicate rows	768 rows
– 3 outlier rows	765 rows
+ imputation applied	765 rows
✓ Final clean dataset	754 rows

Z-SCORE VISUALISATION



Correlation *Matrix.*



KEY OBSERVATIONS

STRONGEST POSITIVE

$r = +0.99$

Y1 HeatingLoad ↔ Y2 CoolingLoad — both outputs driven by identical parameters.

STRONGEST NEGATIVE

$r = -0.99$

X1 Compactness ↔ X2 Surface Area — geometric identity: compact = less surface.

GLAZING → HEATING

$r = +0.81$

Strongest input–output correlation. Consistent with RF importance: 93/100.

ORIENTATION → LOADS

$r \approx 0.01$

Confirms RF finding: X6 orientation explains essentially zero variance in Y1/Y2.

Duplicates, Bias & Consistency.

REPEATED DATA CHECK

Dup.idx	Building	Glazing	Orient.	Cause	Action
Row 10	B1 Cubic 7m	0%	North	Data export error	Removed
Row 45	B4 Plus 7m	10%	East	Data export error	Removed
Row 120	B7 Cubic 3.5m	0%	South	Data export error	Removed

3 duplicates removed · 768 → 765 rows

BIAS CHECK – ORIENTATION

Orient.	Mean HL	Δ from N	Bar
North	20.4	—	
East	20.47	+0.07	
South	20.52	+0.12	
West	20.58	+0.18	

Max diff: 0.18 kWh/m² · RF score: 0/100
Orientation introduces negligible variation. No bias detected.

DATA CONSISTENCY CHECK

Check	Result
X1 within [0.62, 0.98]	✓ PASS
X5 only {3.5, 7.0}	✓ PASS
X6 only {2,3,4,5}	✓ PASS
X7 only {0,0.1,0.25,0.4}	✓ PASS
X8 only {0,1,2,3,4}	✓ PASS
Y1 all positive	✓ PASS
Y2 all positive	✓ PASS
Expected row count (754)	✓ PASS

8 / 8 checks passed · dataset fully consistent